

pandas1

September 22, 2025

0.0.1 Pandas

- pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.
- Unlike numpy Pandas can work with heterogenous data.

```
[1]: import pandas as pd
```

```
[2]: df = pd.read_csv('mckinsey.csv')
```

```
[3]: df
```

```
[3]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	Africa	62.351	706.157306
1700	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298

[1704 rows x 6 columns]

```
[4]: type(df)
```

```
[4]: pandas.core.frame.DataFrame
```

```
[6]: type(df['country'])
```

```
[6]: pandas.core.series.Series
```

```
[8]: # df['country']
```

```
[10]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1704 entries, 0 to 1703
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   country         1704 non-null   object
1   year            1704 non-null   int64
2   population       1704 non-null   int64
3   continent        1704 non-null   object
4   life_exp        1704 non-null   float64
5   gdp_cap         1704 non-null   float64
dtypes: float64(2), int64(2), object(2)
memory usage: 80.0+ KB

```

```
[11]: df
```

```

[11]:
      country  year  population  continent  life_exp  gdp_cap
0  Afghanistan  1952      8425333      Asia    28.801  779.445314
1  Afghanistan  1957      9240934      Asia    30.332  820.853030
2  Afghanistan  1962     10267083      Asia    31.997  853.100710
3  Afghanistan  1967     11537966      Asia    34.020  836.197138
4  Afghanistan  1972     13079460      Asia    36.088  739.981106
...
1699  Zimbabwe  1987      9216418  Africa    62.351  706.157306
1700  Zimbabwe  1992     10704340  Africa    60.377  693.420786
1701  Zimbabwe  1997     11404948  Africa    46.809  792.449960
1702  Zimbabwe  2002     11926563  Africa    39.989  672.038623
1703  Zimbabwe  2007     12311143  Africa    43.487  469.709298

```

[1704 rows x 6 columns]

```
[17]: df.head() # Top 5 rows
```

```

[17]:
      country  year  population  continent  life_exp  gdp_cap
0  Afghanistan  1952      8425333      Asia    28.801  779.445314
1  Afghanistan  1957      9240934      Asia    30.332  820.853030
2  Afghanistan  1962     10267083      Asia    31.997  853.100710
3  Afghanistan  1967     11537966      Asia    34.020  836.197138
4  Afghanistan  1972     13079460      Asia    36.088  739.981106

```

```
[14]: df.head(10)
```

```

[14]:
      country  year  population  continent  life_exp  gdp_cap
0  Afghanistan  1952      8425333      Asia    28.801  779.445314
1  Afghanistan  1957      9240934      Asia    30.332  820.853030
2  Afghanistan  1962     10267083      Asia    31.997  853.100710
3  Afghanistan  1967     11537966      Asia    34.020  836.197138

```

4	Afghanistan	1972	13079460	Asia	36.088	739.981106
5	Afghanistan	1977	14880372	Asia	38.438	786.113360
6	Afghanistan	1982	12881816	Asia	39.854	978.011439
7	Afghanistan	1987	13867957	Asia	40.822	852.395945
8	Afghanistan	1992	16317921	Asia	41.674	649.341395
9	Afghanistan	1997	22227415	Asia	41.763	635.341351

```
[16]: df.tail(2)
```

```
[16]:      country  year  population  continent  life_exp  gdp_cap
1702  Zimbabwe  2002    11926563    Africa    39.989   672.038623
1703  Zimbabwe  2007    12311143    Africa    43.487   469.709298
```

```
[18]: df.shape
```

```
[18]: (1704, 6)
```

```
[ ]:
```

0.0.2 Some basic operations on columns

- Add a column
- Delete a column
- Rename a column

```
[19]: df.columns
```

```
[19]: Index(['country', 'year', 'population', 'continent', 'life_exp', 'gdp_cap'],
      dtype='object')
```

```
[20]: df.keys()
```

```
[20]: Index(['country', 'year', 'population', 'continent', 'life_exp', 'gdp_cap'],
      dtype='object')
```

```
[22]: df.head()
```

0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106

```
[23]: df['country']
```

```
[23]: 0      Afghanistan
      1      Afghanistan
      2      Afghanistan
      3      Afghanistan
      4      Afghanistan
      ...
      1699     Zimbabwe
      1700     Zimbabwe
      1701     Zimbabwe
      1702     Zimbabwe
      1703     Zimbabwe
      Name: country, Length: 1704, dtype: object
```

```
[25]: df['country'].head()
```

```
[25]: 0      Afghanistan
      1      Afghanistan
      2      Afghanistan
      3      Afghanistan
      4      Afghanistan
      Name: country, dtype: object
```

```
[28]: # Accessing multiple columns
```

```
[30]: df[['country', 'population', 'gdp_cap']]
```

```
[30]:
```

	country	population	gdp_cap
0	Afghanistan	8425333	779.445314
1	Afghanistan	9240934	820.853030
2	Afghanistan	10267083	853.100710
3	Afghanistan	11537966	836.197138
4	Afghanistan	13079460	739.981106
...	...	...	...
1699	Zimbabwe	9216418	706.157306
1700	Zimbabwe	10704340	693.420786
1701	Zimbabwe	11404948	792.449960
1702	Zimbabwe	11926563	672.038623
1703	Zimbabwe	12311143	469.709298

```
[1704 rows x 3 columns]
```

```
[32]: df['country'].unique()
```

```
[32]: array(['Afghanistan', 'Albania', 'Algeria', 'Angola', 'Argentina',
        'Australia', 'Austria', 'Bahrain', 'Bangladesh', 'Belgium',
        'Benin', 'Bolivia', 'Bosnia and Herzegovina', 'Botswana', 'Brazil',
        'Bulgaria', 'Burkina Faso', 'Burundi', 'Cambodia', 'Cameroon',
```

```
'Canada', 'Central African Republic', 'Chad', 'Chile', 'China',
'Colombia', 'Comoros', 'Congo, Dem. Rep.', 'Congo, Rep.',
'Costa Rica', 'Cote d'Ivoire', 'Croatia', 'Cuba', 'Czech Republic',
'Denmark', 'Djibouti', 'Dominican Republic', 'Ecuador', 'Egypt',
'El Salvador', 'Equatorial Guinea', 'Eritrea', 'Ethiopia',
'Finland', 'France', 'Gabon', 'Gambia', 'Germany', 'Ghana',
'Greece', 'Guatemala', 'Guinea', 'Guinea-Bissau', 'Haiti',
'Honduras', 'Hong Kong, China', 'Hungary', 'Iceland', 'India',
'Indonesia', 'Iran', 'Iraq', 'Ireland', 'Israel', 'Italy',
'Jamaica', 'Japan', 'Jordan', 'Kenya', 'Korea, Dem. Rep.',
'Korea, Rep.', 'Kuwait', 'Lebanon', 'Lesotho', 'Liberia', 'Libya',
'Madagascar', 'Malawi', 'Malaysia', 'Mali', 'Mauritania',
'Mauritius', 'Mexico', 'Mongolia', 'Montenegro', 'Morocco',
'Mozambique', 'Myanmar', 'Namibia', 'Nepal', 'Netherlands',
'New Zealand', 'Nicaragua', 'Niger', 'Nigeria', 'Norway', 'Oman',
'Pakistan', 'Panama', 'Paraguay', 'Peru', 'Philippines', 'Poland',
'Portugal', 'Puerto Rico', 'Reunion', 'Romania', 'Rwanda',
'Sao Tome and Principe', 'Saudi Arabia', 'Senegal', 'Serbia',
'Sierra Leone', 'Singapore', 'Slovak Republic', 'Slovenia',
'Somalia', 'South Africa', 'Spain', 'Sri Lanka', 'Sudan',
'Swaziland', 'Sweden', 'Switzerland', 'Syria', 'Taiwan',
'Tanzania', 'Thailand', 'Togo', 'Trinidad and Tobago', 'Tunisia',
'Turkey', 'Uganda', 'United Kingdom', 'United States', 'Uruguay',
'Venezuela', 'Vietnam', 'West Bank and Gaza', 'Yemen, Rep.',
'Zambia', 'Zimbabwe'], dtype=object)
```

```
[33]: len(df['country'].unique())
```

```
[33]: 142
```

```
[34]: df['country'].nunique()
```

```
[34]: 142
```

```
[47]: # df.value_counts() -> gives all unique values along with the freq of those
      ↪ values:
```

```
[35]: df['country'].value_counts()
```

```
[35]: Afghanistan      12
      Pakistan         12
      New Zealand      12
      Nicaragua        12
      Niger            12
      ..
      Eritrea          12
      Equatorial Guinea 12
```

```

El Salvador      12
Egypt            12
Zimbabwe         12
Name: country, Length: 142, dtype: int64

```

```
[38]: # df.index
```

```
[ ]:
```

```
[39]: # renaming our columns
```

```
[40]: df.head()
```

```
[40]:
```

	country	year	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106

```
[41]: df.rename(columns = {'country' : 'COUNTRY', 'year' : 'YEAR'})
```

```
[41]:
```

	COUNTRY	YEAR	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106
...	...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	Africa	62.351	706.157306
1700	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298

```
[1704 rows x 6 columns]
```

```
[44]: # df
```

```
[45]: df.rename({'country' : 'COUNTRY', 'year' : 'YEAR'}, axis = 1)
```

```
[45]:
```

	COUNTRY	YEAR	population	continent	life_exp	gdp_cap
0	Afghanistan	1952	8425333	Asia	28.801	779.445314
1	Afghanistan	1957	9240934	Asia	30.332	820.853030
2	Afghanistan	1962	10267083	Asia	31.997	853.100710
3	Afghanistan	1967	11537966	Asia	34.020	836.197138
4	Afghanistan	1972	13079460	Asia	36.088	739.981106

...	...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	Africa	62.351	706.157306
1700	Zimbabwe	1992	10704340	Africa	60.377	693.420786
1701	Zimbabwe	1997	11404948	Africa	46.809	792.449960
1702	Zimbabwe	2002	11926563	Africa	39.989	672.038623
1703	Zimbabwe	2007	12311143	Africa	43.487	469.709298

[1704 rows x 6 columns]

```
[48]: # How do we access columns
```

```
[50]: df['country']
```

```
[50]: 0      Afghanistan
      1      Afghanistan
      2      Afghanistan
      3      Afghanistan
      4      Afghanistan
      ...
      1699      Zimbabwe
      1700      Zimbabwe
      1701      Zimbabwe
      1702      Zimbabwe
      1703      Zimbabwe
      Name: country, Length: 1704, dtype: object
```

```
[54]: # df.country
```

```
[53]: df.shape
```

```
[53]: (1704, 6)
```

```
[ ]:
```

0.0.3 Drop a column

```
[58]: df.drop('continent', axis = 1)
```

```
[58]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	62.351	706.157306
1700	Zimbabwe	1992	10704340	60.377	693.420786

1701	Zimbabwe	1997	11404948	46.809	792.449960
1702	Zimbabwe	2002	11926563	39.989	672.038623
1703	Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 5 columns]

```
[60]: df.drop(columns = 'continent')
```

```
[60]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	62.351	706.157306
1700	Zimbabwe	1992	10704340	60.377	693.420786
1701	Zimbabwe	1997	11404948	46.809	792.449960
1702	Zimbabwe	2002	11926563	39.989	672.038623
1703	Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 5 columns]

```
[62]: df.drop(columns = 'continent', inplace = True)
```

```
[64]: df
```

```
[64]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	62.351	706.157306
1700	Zimbabwe	1992	10704340	60.377	693.420786
1701	Zimbabwe	1997	11404948	46.809	792.449960
1702	Zimbabwe	2002	11926563	39.989	672.038623
1703	Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 5 columns]

```
[ ]:
```


0.0.4 Creating new columns

```
[65]: df.head()
```

```
[65]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106

```
[66]: df["year + 7"] = df['year'] + 7
```

```
[68]: df.head()
```

```
[68]:
```

	country	year	population	life_exp	gdp_cap	year + 7
0	Afghanistan	1952	8425333	28.801	779.445314	1959
1	Afghanistan	1957	9240934	30.332	820.853030	1964
2	Afghanistan	1962	10267083	31.997	853.100710	1969
3	Afghanistan	1967	11537966	34.020	836.197138	1974
4	Afghanistan	1972	13079460	36.088	739.981106	1979

```
[70]: df['gdp'] = df['gdp_cap'] * df['population']
```

```
[72]: df.head()
```

```
[72]:
```

	country	year	population	life_exp	gdp_cap	year + 7	gdp
0	Afghanistan	1952	8425333	28.801	779.445314	1959	6.567086e+09
1	Afghanistan	1957	9240934	30.332	820.853030	1964	7.585449e+09
2	Afghanistan	1962	10267083	31.997	853.100710	1969	8.758856e+09
3	Afghanistan	1967	11537966	34.020	836.197138	1974	9.648014e+09
4	Afghanistan	1972	13079460	36.088	739.981106	1979	9.678553e+09

```
[77]: df.shape
```

```
[77]: (1704, 8)
```

```
[74]: df.shape[0]
```

```
[74]: 1704
```

```
[75]: df['own'] = [i for i in range(df.shape[0])]
```

```
[76]: df.head()
```

```
[76]:
```

	country	year	population	life_exp	gdp_cap	year + 7	\
0	Afghanistan	1952	8425333	28.801	779.445314	1959	

1	Afghanistan	1957	9240934	30.332	820.853030	1964
2	Afghanistan	1962	10267083	31.997	853.100710	1969
3	Afghanistan	1967	11537966	34.020	836.197138	1974
4	Afghanistan	1972	13079460	36.088	739.981106	1979

		gdp	own
0	6.567086e+09	0	
1	7.585449e+09	1	
2	8.758856e+09	2	
3	9.648014e+09	3	
4	9.678553e+09	4	

```
[78]: df.drop(columns = ['own', 'gdp', 'year + 7'], inplace = True)
```

```
[79]: df.head()
```

```
[79]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106

```
[ ]:
```

0.0.5 Some operations on rows

```
[80]: df.index
```

```
[80]: RangeIndex(start=0, stop=1704, step=1)
```

```
[81]: df
```

```
[81]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	62.351	706.157306
1700	Zimbabwe	1992	10704340	60.377	693.420786
1701	Zimbabwe	1997	11404948	46.809	792.449960
1702	Zimbabwe	2002	11926563	39.989	672.038623
1703	Zimbabwe	2007	12311143	43.487	469.709298

```
[1704 rows x 5 columns]
```

```
[84]: df.shape[0] + 1
```

```
[84]: 1705
```

```
[112]: df.index = list(range(1, df.shape[0] + 1)) # create a list of indices
```

```
[113]: df
```

```
[113]:
```

	country	year	population	life_exp	gdp_cap
1	Afghanistan	1952	8425333	28.801	779.445314
2	Afghanistan	1957	9240934	30.332	820.853030
3	Afghanistan	1962	10267083	31.997	853.100710
4	Afghanistan	1967	11537966	34.020	836.197138
5	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1700	Zimbabwe	1987	9216418	62.351	706.157306
1701	Zimbabwe	1992	10704340	60.377	693.420786
1702	Zimbabwe	1997	11404948	46.809	792.449960
1703	Zimbabwe	2002	11926563	39.989	672.038623
1704	Zimbabwe	2007	12311143	43.487	469.709298

```
[1704 rows x 5 columns]
```

```
[88]: # explicit -> user defined  
# implicit -> python style defined
```

```
[94]: # accessing using index -> column
```

```
[95]: # df[1]
```

```
[ ]:
```

```
[98]: # slicing -> returning rows
```

```
[97]: df[:4]
```

```
[97]:
```

	country	year	population	life_exp	gdp_cap
1	Afghanistan	1952	8425333	28.801	779.445314
2	Afghanistan	1957	9240934	30.332	820.853030
3	Afghanistan	1962	10267083	31.997	853.100710
4	Afghanistan	1967	11537966	34.020	836.197138

```
[101]: import numpy as np
```

```
[102]: df.index = np.arange(1, df.shape[0] + 1, dtype='float')
```

```
[103]: df
```

```
[103]:
```

	country	year	population	life_exp	gdp_cap
1.0	Afghanistan	1952	8425333	28.801	779.445314
2.0	Afghanistan	1957	9240934	30.332	820.853030
3.0	Afghanistan	1962	10267083	31.997	853.100710
4.0	Afghanistan	1967	11537966	34.020	836.197138
5.0	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1700.0	Zimbabwe	1987	9216418	62.351	706.157306
1701.0	Zimbabwe	1992	10704340	60.377	693.420786
1702.0	Zimbabwe	1997	11404948	46.809	792.449960
1703.0	Zimbabwe	2002	11926563	39.989	672.038623
1704.0	Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 5 columns]

```
[104]: sample = df.head()
```

```
[106]: sample.index = ['a', 'b', 'c', 'd', 'e']
```

```
[107]: sample
```

```
[107]:
```

	country	year	population	life_exp	gdp_cap
a	Afghanistan	1952	8425333	28.801	779.445314
b	Afghanistan	1957	9240934	30.332	820.853030
c	Afghanistan	1962	10267083	31.997	853.100710
d	Afghanistan	1967	11537966	34.020	836.197138
e	Afghanistan	1972	13079460	36.088	739.981106

```
[109]: df.index[0] # returns explicit index corresponding to implicit
```

```
[109]: 1.0
```

```
[ ]:
```

```
[114]: ser = df['country']
```

```
[116]: ser.head(20)
```

```
[116]:
```

1	Afghanistan
2	Afghanistan
3	Afghanistan
4	Afghanistan
5	Afghanistan
6	Afghanistan
7	Afghanistan
8	Afghanistan
9	Afghanistan

```

10    Afghanistan
11    Afghanistan
12    Afghanistan
13        Albania
14        Albania
15        Albania
16        Albania
17        Albania
18        Albania
19        Albania
20        Albania
Name: country, dtype: object

```

```

[121]: # slicing -> implicit indices
ser[5:13]

```

```

[121]: 6    Afghanistan
7    Afghanistan
8    Afghanistan
9    Afghanistan
10   Afghanistan
11   Afghanistan
12   Afghanistan
13        Albania
Name: country, dtype: object

```

```

[117]: ser[12]

```

```

[117]: 'Afghanistan'

```

```

[118]: ser[13]

```

```

[118]: 'Albania'

```

```

[ ]:

```

```

[122]: ## iloc -> implicit
      ## loc -> explicit

```

```

[125]: df.head()

```

```

[125]:
   country  year  population  life_exp  gdp_cap
1  Afghanistan  1952      8425333    28.801  779.445314
2  Afghanistan  1957      9240934    30.332  820.853030
3  Afghanistan  1962     10267083    31.997  853.100710
4  Afghanistan  1967     11537966    34.020  836.197138
5  Afghanistan  1972     13079460    36.088  739.981106

```

```
[130]: df.iloc[0]
```

```
[130]: country      Afghanistan
      year          1952
      population    8425333
      life_exp      28.801
      gdp_cap       779.445314
      Name: 1, dtype: object
```

```
[129]: # iloc -> implicit index

      df.iloc[1]
```

```
[129]: country      Afghanistan
      year          1957
      population    9240934
      life_exp      30.332
      gdp_cap       820.85303
      Name: 2, dtype: object
```

```
[ ]:
```

```
[128]: # df.loc[0]
```

```
[124]: # explicit index value

      df.loc[1]
```

```
[124]: country      Afghanistan
      year          1952
      population    8425333
      life_exp      28.801
      gdp_cap       779.445314
      Name: 1, dtype: object
```

```
[ ]:
```

```
[131]: df.head()
```

```
[131]:
```

	country	year	population	life_exp	gdp_cap
1	Afghanistan	1952	8425333	28.801	779.445314
2	Afghanistan	1957	9240934	30.332	820.853030
3	Afghanistan	1962	10267083	31.997	853.100710
4	Afghanistan	1967	11537966	34.020	836.197138
5	Afghanistan	1972	13079460	36.088	739.981106

```
[134]: # In iloc start is inclusive but end is exclusive

      df.iloc[1:4]
```

```
[134]:      country  year  population  life_exp  gdp_cap
      2  Afghanistan  1957      9240934    30.332  820.853030
      3  Afghanistan  1962     10267083    31.997  853.100710
      4  Afghanistan  1967     11537966    34.020  836.197138
```

```
[ ]:
```

```
[135]: # In loc both start and end are inclusive

df.loc[1:4]
```

```
[135]:      country  year  population  life_exp  gdp_cap
      1  Afghanistan  1952      8425333    28.801  779.445314
      2  Afghanistan  1957      9240934    30.332  820.853030
      3  Afghanistan  1962     10267083    31.997  853.100710
      4  Afghanistan  1967     11537966    34.020  836.197138
```

```
[138]: df.head(10)
```

```
[138]:      country  year  population  life_exp  gdp_cap
      1  Afghanistan  1952      8425333    28.801  779.445314
      2  Afghanistan  1957      9240934    30.332  820.853030
      3  Afghanistan  1962     10267083    31.997  853.100710
      4  Afghanistan  1967     11537966    34.020  836.197138
      5  Afghanistan  1972     13079460    36.088  739.981106
      6  Afghanistan  1977     14880372    38.438  786.113360
      7  Afghanistan  1982     12881816    39.854  978.011439
      8  Afghanistan  1987     13867957    40.822  852.395945
      9  Afghanistan  1992     16317921    41.674  649.341395
     10  Afghanistan  1997     22227415    41.763  635.341351
```

```
[139]: df.loc[9::-3]
```

```
[139]:      country  year  population  life_exp  gdp_cap
      9  Afghanistan  1992     16317921    41.674  649.341395
      6  Afghanistan  1977     14880372    38.438  786.113360
      3  Afghanistan  1962     10267083    31.997  853.100710
```

```
[141]: temp = df.set_index("country")
```

```
[142]: temp
```

```
[142]:      year  population  life_exp  gdp_cap
country
Afghanistan  1952      8425333    28.801  779.445314
Afghanistan  1957      9240934    30.332  820.853030
Afghanistan  1962     10267083    31.997  853.100710
```

Afghanistan	1967	11537966	34.020	836.197138
Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...
Zimbabwe	1987	9216418	62.351	706.157306
Zimbabwe	1992	10704340	60.377	693.420786
Zimbabwe	1997	11404948	46.809	792.449960
Zimbabwe	2002	11926563	39.989	672.038623
Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 4 columns]

```
[145]: df.reset_index(drop = True)
```

```
[145]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	62.351	706.157306
1700	Zimbabwe	1992	10704340	60.377	693.420786
1701	Zimbabwe	1997	11404948	46.809	792.449960
1702	Zimbabwe	2002	11926563	39.989	672.038623
1703	Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 5 columns]

```
[148]: df.reset_index(drop = True, inplace=True)
```

```
[149]: df
```

```
[149]:
```

	country	year	population	life_exp	gdp_cap
0	Afghanistan	1952	8425333	28.801	779.445314
1	Afghanistan	1957	9240934	30.332	820.853030
2	Afghanistan	1962	10267083	31.997	853.100710
3	Afghanistan	1967	11537966	34.020	836.197138
4	Afghanistan	1972	13079460	36.088	739.981106
...	...	...	...	...	...
1699	Zimbabwe	1987	9216418	62.351	706.157306
1700	Zimbabwe	1992	10704340	60.377	693.420786
1701	Zimbabwe	1997	11404948	46.809	792.449960
1702	Zimbabwe	2002	11926563	39.989	672.038623
1703	Zimbabwe	2007	12311143	43.487	469.709298

[1704 rows x 5 columns]


```
[150]: # read about concat() method
```

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